

Ruminal Fermentation and Fill Change with Season in an Arctic Grazer: Responses to Hyperphagia and Hypophagia in Muskoxen (*Ovibos moschatus*)

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ABSTRACT

We studied castrated adult muskoxen fed a standard diet of grass hay and supplement throughout the year to determine seasonal changes in digesta passage, fill, and fermentation without the confounding effects of reproductive demands or changes in food quality. Although food intake increased by 74% between spring and autumn, mean retention times of fluid and particulate digesta markers were maintained between seasons in both the rumen (9–13 h) and the intestines (27–37 h). The rumen contained 84.5% of digesta and accounted for 79% of dry matter digestion in the whole digestive tract. Ruminal fluid space and whole-gut digesta fill increased by 31%–34%, while ruminal rates of in situ degradation increased by more than 100% between spring and autumn for cellulose and hemicellulose. Hyperphagia in autumn was accompanied by increased bacterial counts in ruminal fluid (30%), declines in ruminal pH, and increases in the concentration of fermentation acids (16%) when compared with spring hypophagia. Consumption of fresh hay and supplement increased the concentrations of acids most markedly during winter and spring when bacterial counts were low. Low food intakes in winter and spring may limit the microbial population, whereas hyperphagia in autumn may foster a much more active microflora that requires consistent supplies of substrate. Plasticity of fill and fermentation in muskoxen minimizes winter costs and maximizes nutrients

and energy gained from coarse forages in small home ranges throughout the year.

Introduction

Seasonal fluctuations of food intake in arctic ungulates such as muskoxen (*Ovibos moschatus*) and reindeer (*Rangifer tarandus*) accompany dramatic changes in photoperiod and plant growth at high latitudes. Winter food intakes are influenced by snow and ice that impair mobility and limit access to senesced or dormant forages (Nellemann 1997). Consumption of growing plants from spring to autumn must therefore support demands for reproduction as well the deposition of fat and lean body reserves for the ensuing winter (Forchhammer and Boomsma 1995; Barboza and Bowyer 2001). Seasonal patterns of food intake are also apparent in captive animals as metabolic demands for energy and nutrients to maintain or gain body mass still change with season when food is available ad lib. throughout the year (White et al. 1987; Barboza et al. 2004a). Endogenous cycles of appetite and digestive function therefore establish the range of functional response to changes in quantity and quality of food. These physiological patterns allow ungulates to respond to food supplies that vary with environmental conditions in a predictable fashion between seasons and in an unpredictable fashion between years throughout their life cycle (Barboza and Hume 2006).

Elevated food intakes (hyperphagia) increase the demand for gut capacity to accommodate the greater load of food and digesta. Hyperphagia may also result in faster flow of digesta as meals increase in size or frequency. Faster passage of digesta may, however, reduce the time available for digestion and absorption if the capacity of the gut does not increase in concert with load of digesta (Sibly 1981). Animals that use the energy from digestion of plant cell walls are least suited to increases in passage rate because microbial degradation of fiber is typically slow (Alexander 1996). The foregut of ruminants is poorly suited to fluctuations in flow and best suited to the maintenance of steady states of mixing and efflux (Hume 1989). Grazing ruminants, such as muskoxen, that consume bulky fibrous foods (Klein and Bay 1995; Schaefer and Messier 1996; Larter and Nagy 1997) are therefore poorly suited to hyperphagia. Nonetheless, muskoxen fed grass hay and supplement sustain

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hyperphagia without a loss in digestive efficiency (Peltier et al. 2003). Digestive efficiencies for dry matter (DM; 0.70 g g^{-1}), energy (0.69 kJ kJ^{-1}), cellulose (0.74 g g^{-1}), and nitrogen (0.63 g g^{-1}) over the whole gut did not decline when intakes of DM, energy cellulose, and organic N increased by 74%, 67%, 59%, and 135%, respectively, between spring and autumn (Peltier et al. 2003).

Muskoxen can process more of the same substrates each day during autumn than in spring (Peltier 2003); that is, digestive processes involving endogenous secretions as well as exogenous enzymes from microbes are sustained as substrate load increases during hyperphagia. Endogenous systems for secretion, digestion, and absorption in the gastric stomach, pancreas, small intestine, and liver can respond rapidly to high-ingestive loads in small mammals and birds (Karasov and Hume 1997). Digestive fermentations in the foregut and the hindgut are much more complex than these endogenous systems because fermentations involve a diverse microbiota (Mackie et al. 2000) interacting with the physical and chemical characteristics of digesta (Van Soest 1994; Cork et al. 1999) and its many substrates (Flint and Forsberg 1995; Russell and Rychlik 2001). Seasonal changes in the digestive fermentations of northern ungulates have been related to changes in the quality or digestibility of the diet (White and Staaland 1983; Orpin et al. 1985; Jiang and Hudson 1996; Mathiesen et al. 2000). We do not know whether ruminal fermentation changes with seasonal hyperphagia when the confounding effect of food quality is removed. In domestic sheep (*Ovis aries*), seasonal variation in food intakes are mainly associated with changes in food quality (Iason et al. 2000). Increased food intake of the same diet in domestic ruminants increases ruminal fill but does not alter digestive efficiency or distribution of bacteria among digesta phases (Rodríguez et al. 2003). If northern ruminants are similar to their domestic counterparts, seasonal hyperphagia on the same diet would affect only the fill and flow of digesta without altering microbial activity.

Slow passage of digesta that often accompany low food intakes (hypophagia) should provide abundant time for slow digestive processes such as the microbial degradation of fiber. Low food intakes in winter should favor fiber digestion in northern ungulates such as muskoxen. Fiber digestibility of a common winter diet is greater in muskoxen than in cattle (*Bos taurus*) probably because food intakes and passage rates are lower in muskoxen than in cattle (Adamczewski et al. 1994a, 1994b). Degradation of fiber does not increase in muskoxen between autumn and winter; on the contrary, cellulose digestibilities decline from autumn (0.76 g g^{-1}) to winter (0.65 g g^{-1} ; Peltier et al. 2003). Starvation can dramatically suppress microbial numbers and activities in ruminants (Aagnes 1995; Dehority 2003), but we do not know the effect of seasonal hypophagia on digestive function when food is available ad lib.

We pose two questions in this article: How do muskoxen sustain hyperphagia in autumn? And what are the digestive

consequences of hypophagia in winter and spring? We conducted two experiments on muskoxen in winter, spring, and autumn. We used castrated adult males to exclude any effects of reproductive demands on food intake and digestive function. First, we describe the distribution of digesta fill and the apparent digestion of dietary components along the digestive tract of animals fed grass hay during late winter. Second, we test two predictions on seasonal responses to food intake: digesta fill and passage increase in autumn hyperphagia; and microbial number and activity in the rumen increase with hyperphagia and decrease with hypophagia. We present new data on digesta passage and microbial activity from an experiment on seasonal balance of muskoxen (Peltier et al. 2003). This article follows a brief preliminary report on seasonal microbial activities in muskoxen (Barboza et al. 2004b).

Material and Methods

Muskoxen were held and studied at the R. G. White Large Animal Research Station (Fairbanks, AK; 65°N , 146°W) under protocol 00-003, with approval of the Institutional Animal Care and Use Committee, University of Alaska, Fairbanks. Seasons at this latitude are described as follows: early winter = October–January, late winter = February–April, spring and summer = May–July, and autumn = August and September.

Segmental Capacity and Function

We used five castrated males (9.9–6.9 yr) in late winter (April 12–24, 2002). Animals were held in a common pen and fed grass hay (*Bromus* sp.) ad lib. with snow as their principal source of water. Body mass was recorded to the nearest 0.5 kg (model 703 scale, Tru-Test, San Antonio, TX). A pelleted mixture of cereal grains, roughage, and premixes of vitamins and minerals was provided as a supplement at $35 \text{ g kg}^{-0.75}$ body mass each week (M Ration, Alaska Pet and Garden, Anchorage, AK). Supplement was withheld for 5–7 d before necropsy to ensure that residues in the digestive tract were primarily associated with grass hay. Consumption of natural forage was minimal during this period because pastures were covered with snow and had not begun spring growth.

Animals were immobilized with a mixture of tiletamine and zolazepam (3 mg kg^{-1} ; Telazol 100 mg mL^{-1} , Fort Dodge Animal Health, Fort Dodge, IA) and then killed by barbiturate overdose ($10\text{--}18 \text{ mL}$ containing 390 mg mL^{-1} pentobarbital sodium and 50 mg mL^{-1} phenytoin sodium; Euthasol, Delmarva Laboratories, Midlothian, VA). Dissection of the digestive tract began immediately after cardiac arrest and the loss of the corneal reflex. The digestive tract was ligated to isolate the ruminoreticulum, omasum, abomasum (gastric stomach), small intestine (pylorus to ileocecal juncture), cecum, proximal colon (cecal juncture to the spiral colon), distal colon, and the rectum (terminal expansion of the colon where fecal pellets

accumulated). Digesta contents were measured as the difference in mass between full and empty segments. Samples of digesta were stored at -20°C for analysis.

Seasonal Digestive Function

We used eight castrated adults (3.8–8.8 yr) that were surgically prepared with cannulae in the dorsal rumen ($n = 8$) and at the duodenum ($n = 6$) within 0.5 m of the pyloric junction. Flexible ruminal cannulae (#7C, Bar Diamond, Parma, ID) were 5-cm internal diameter, whereas duodenal cannulae were stainless steel with 1-cm internal diameter (custom fabrication with an internal flange). All animals were completely recovered from surgery and eating normally for 1.5 mo before the start of experiments. The number of patent duodenal cannulae declined to four during the experiment because regrowth of the mucosa permanently occluded the cannulae in two animals after 6 mo.

Muskoxen were trained to frequent handling and to mild physical restraint in a squeeze chute for weighing, dosing with markers, and digesta sampling without chemical immobilants or sedation. Animals were housed in individual pens in the spring (May and June 2000) and autumn (August and September 2000) but divided into two groups and held in one of two large pens (0.139 or 0.184 ha) during winter (October 2000–March 2001). Fresh water or snow was provided with grass hay (*Bromus* sp.) ad lib. throughout the year. Animals were fed either a basic supplement or a high-protein supplement each day at $5 \text{ g kg}^{-0.75} \text{ d}^{-1}$ in spring and in winter and at $5.7 \text{ g kg}^{-0.75} \text{ d}^{-1}$ during autumn. Both supplements included protein from corn (7%–22% as fed), barley (7%–22% as fed), and alfalfa (19% as fed), whereas fish meal (20% as fed) was included as an additional source of N in the high-protein supplement. Variation in diets was minimized by producing a single batch of each supplement for the entire experiment (Alaska Pet and Garden, Anchorage, AK) and by using a single consignment of hay within each season. Muskoxen were ranked by size and assigned to one of two groups for the duration of the experiment over three seasons: spring (May and June), autumn (August and September), and winter (February and March). Each seasonal period included two trials of 4 wk in a crossover design. Experimental trials consisted of 2 wk of acclimation to a supplement followed by 10–14 d during which food, feces, and ruminal digesta were collected for measures of intake and digestive function.

Food intake was measured in spring and autumn by weighing food bins before and after refilling with fresh hay each morning. Representative samples of hay and supplement were collected in each period as described by (Peltier et al. 2003). Samples of spontaneously voided feces were collected each morning and at various intervals during daylight hours for measures of digestibility and digesta passage. Samples of ruminal digesta were collected by restraining animals in a holding chute and by

inserting a 4×40 -cm flexible tube into the rumen to withdraw a column of fluid and particulate digesta via the cannula.

Ruminal digesta were collected before and after (6 h) animals were fed supplement and fresh hay. Digesta pH was measured with narrow-range papers (± 0.1 pH unit) within 30 s of removal from the rumen. Ruminal fluid was extruded through several layers of gauze to sample for bacterial counts and for analysis of fermentation acids. Samples for bacterial counts were collected into a tared vial with an equal volume of 10% v/v formalin buffered at pH 7. Formolized samples were stored at room temperature for counting. Samples for analysis of fermentation acids were collected into a tared vial with 0.5 mL saturated mercuric chloride (HgCl_2) to arrest all enzyme activity before storing the samples at -20°C .

Digesta passage was measured with two markers: Co-EDTA for aqueous fluids and Yb-fiber for particles. Co-EDTA was prepared as the sodium salt (Uden et al. 1980). Particles were prepared from muskox feces by wet sieving with water through 1-mm mesh onto 0.5-mm mesh. Fiber was extracted from particles in neutral detergent solution (Van Soest et al. 1991), rinsed thoroughly, and bound with Yb-acetate tetrahydrate at $100 \text{ mg g}^{-1} \text{ DM}$ in 0.01 M acetic acid (Ellis and Beever 1984; de Vega et al. 1998). Subsequent rinses with 0.1 M acetic acid removed weakly bound Yb. Wet sieving with water onto $75\text{-}\mu\text{m}$ mesh removed excess acid and ultrafine material. Labeled particles were dried at 55°C for dosing.

Co-EDTA was dosed in solution (10 g L^{-1} water) at the ruminal or duodenal cannulae. Particles of Yb-fiber were loaded into gel capsules for doses at each cannula (Torpac Gel Cap, East Hanover, NJ). Single doses of the fluid marker ($145.3 \pm 2.9 \text{ mg Co}$) and the particulate marker ($11.5 \pm 1.2 \text{ mg Yb}$) were administered at the duodenal cannula between 8:00 a.m. and 10:00 a.m. Ruminal doses of $294.8 \pm 0.8 \text{ mg Co}$ and $40.7 \pm 2.9 \text{ mg Yb}$ were also administered in the morning after 7–8 d from the duodenal dose. Fecal samples were collected to measure marker concentrations before each dose at 4, 9, 15, 24, and 30 h after dosing and at daily intervals until the end of each trial. Ruminal samples were collected before dosing and at 4, 24, and 48 h after the dose. Samples of feces and ruminal digesta were stored at -20°C for analysis.

Ruminal degradation rates were measured in situ by incubating foods in individual permeable bags that were suspended within the rumen. The effects of mechanical degradation are excluded from these measures because the bags were not subjected to rumination and chewing. We incubated both components of the diet (hay and supplement) in each animal during every seasonal trial. Ruminal rates of degradation were therefore measured over a range of dietary qualities that varied in the proportions of structural and nonstructural carbohydrate as well as protein.

Hay was ground before ruminal incubation (2-mm screen, Wiley mill, A. H. Thomas, Philadelphia, PA) to provide a range of particle sizes similar to that of feces. Supplements were in-

cubated without grinding because the pelleted food readily disaggregated when wet. Duplicate samples of hay (5 g) and supplements (10 g) were weighed into N-free polyester bags (5 cm × 10 cm, 50-μm pore size; Ankom Technology, Macedon, NY) for comparison with unfilled control bags that remained in the rumen for 48 h. All bags were inserted into the rumen in the morning before animals were fed supplement and fresh hay. Bags with supplement were withdrawn in duplicate after 6, 24, and 48 h, whereas bags containing hay were withdrawn after 24 and 48 h. Bags were dried at 55°C to constant mass to determine the residual dry mass of the contents after correcting for mass gained by empty controls. Residues of hay and each supplement were pooled, respectively, in each time point within each period to provide representative samples for chemical analysis.

Chemical Analyses

Food, feces, and digesta from both experiments were dried at 55°C to a constant mass to determine DM content. Dried samples were ground through a #20 (1.25 mm) screen in a Wiley mill before analysis. Fiber and lignin was analyzed by the methods of Van Soest et al. (1991): neutral detergent fiber (NDF) was extracted with Na₂SO₃ and amylase; lignin was extracted in H₂SO₄ without permanganate. Hemicellulose content was calculated as the difference between sequential extractions of NDF and acid detergent fiber (ADF). Cellulose content was the difference between sequential extractions of ADF and lignin. Carbon, nitrogen, and sulfur were analyzed with an elemental analyzer (CNS2000, LECO, St. Joseph, MI). Dietary N was converted to crude protein (CP) by assuming that 100 g CP contained 16 g N (Robbins 1993). Ash was determined by combustion in a muffle furnace at 500°C and subtracted from DM to measure organic matter (OM). Residual OM was calculated by subtracting CP and NDF from OM. Residual OM approximates nonstructural carbohydrates such as starch because lipid concentrations are typically less than 1% DM of hay and less than 3% DM of supplement (Rombach et al. 2002a).

Feces and ruminal digesta were prepared for marker analysis by digestion in a mixture of perchloric, nitric, and sulfuric acids (Barboza et al. 2003; Trumble et al. 2003). Digests were diluted with distilled, deionized water, and assayed by directly coupled plasma spectrometry (Iris DCP, Thermo Elemental, Cheshire, UK) against certified standards.

We used a bright-line counting chamber (model 1492, Hauser Scientific, Horsham, PA) with a light microscope to count ruminal bacteria in three fields of 0.25 nL for three aliquots of each sample. Coefficients of variation were 8% for counts within aliquots and within samples. Bacterial counts were expressed on the basis of ruminal fluid (mL⁻¹) after correcting for dilution with formalin.

Ruminal fermentation acids were prepared by centrifugation

(5,000 g) of 3.3 g rumen fluid with 0.7 g of 25% w/v metaphosphoric acid. Digesta fluids were analyzed for short chain fatty acids (SCFA) by gas chromatography (HP model 5890, Agilent, Wilmington, DE) on a capillary column (30 m × 0.32 mm i.d., 1-μm film, ZB-FFAP column, Phenomenex, Torrance, CA) with a flame-ionization detector. Injector temperature was 200°C, the detector temperature was 250°C, and the carrier gas was helium. SCFA were eluted in a sequence of oven temperatures as follows: start at 100°C for 1 min; ramp to 140°C for 2 min; ramp to 200°C for 7.5 min; final plateau at 200°C for 5 min. We used crotonic acid as the internal standard to quantify acetate, propionate, n-butyrate, isobutyrate, valerate, isovalerate, and caproate against the retention times of certified standards. Elution profiles were analyzed with HP 3365 Series I Chemstation (Agilent). Concentrations of individual SCFA were expressed on the basis of ruminal fluid (mM) after correcting for the addition of mercuric chloride and metaphosphoric acid.

Calculations

The digestible fraction of DM in feces or digesta from both experiments was calculated against lignin as an indigestible marker:

$$\text{digestible fraction} = \frac{1 - (\text{diet lignin/diet nutrient})}{(\text{digesta or fecal lignin/digesta or fecal nutrient})}$$

The use of lignin as an indigestible marker assumes that this phenolic complex is completely recovered in a similar form from food, digesta, and feces in each season (Fahey and Jung 1983; Susmel and Stefanon 1993). We could not validate total recovery of lignin in feces from muskoxen because total collection was not feasible for these large animals. We minimized variation in the form of dietary lignin by feeding muskoxen one type of forage, which was the principal source of lignin throughout these studies. Our calculations of segmental digestion were based on lignin in hay because hay accounted for 97% of the DM ingested by muskoxen in spring and autumn (Peltier et al. 2003). Hay prepared from mature temperate grasses is less variable than other forages in the form and composition of lignin (Jung and Fahey 1983). High reproducibility (<10% coefficient of variation) in measures of lignin in food, digesta, and feces were consistent with low variation in the chemical isolation of the marker.

Measures of digesta passage were derived from the concentration of Co and Yb in sequential samples of ruminal digesta or feces. We assumed that each marker labeled a representative pool of digesta that was instantaneously mixed with aqueous fluids (Co) or small indigestible particles 75–500 μm (Yb). The time of excretion of feces was recorded as the median time between collections or as the time at which the defecation was

Table 1: Dry matter composition of grass hay and supplements fed to castrated male muskoxen in segmental and seasonal function experiments

Dietary Component	Segmental Function		Seasonal Function				
	Winter Hay	Supplement	Basic Supplement	Protein Supplement	Spring Hay	Autumn Hay	Winter Hay
OM (g 100 g ⁻¹)	92.5	90.8	92.3	83.3	92.3	93.2	95.7
NDF (g 100 g ⁻¹)	61.6	32.5	24.1	24.0	62.1	59.8	60.1
Hemicellulose (g 100 g ⁻¹)	27.3	15.7	11.4	12.7	28.0	28.0	24.9
ADF (g 100 g ⁻¹)	34.2	16.8	12.6	11.3	34.1	31.8	35.0
Cellulose (g 100 g ⁻¹)	30.0	14.3	10.2	9.8	30.9	29.0	30.7
Lignin (g 100 g ⁻¹)	4.2	2.5	2.5	1.5	3.2	2.7	4.2
CP (g 100 g ⁻¹)	11.0	14.6	13.3	25.5	5.9	9.6	4.5
S (g kg ⁻¹)	1.48	3.68	3.02	4.28	.61	1.19	.65
C (g 100 g ⁻¹)	44.7	42.8	41.9	39.0	44.1	44.7	44.2
Residual OM (g 100 g ⁻¹) ^a	20.0	43.7	55.0	33.8	24.3	23.9	31.4

Note. OM = organic matter; NDF = neutral detergent fiber; ADF = acid detergent fiber; CP = crude protein.

^a Approximation of nonstructural carbohydrate = OM - (NDF + CP).

observed. Ruminal and fecal marker concentrations declined from a maximum after each dose. The natural logarithm of the decline in ruminal marker concentration was fitted to a least squares linear regression against time from dosing. We therefore assumed a simple model of first-order kinetics for both fluid and particulate markers in the rumen. Turnover time (h) of the ruminal marker pool was the negative inverse of the slope of the regression ($-h^{-1}$). The size of the marked fluid pool was calculated by dividing the dose ($\mu\text{g Co}$ or Yb) with the concentration of marker ($\mu\text{g g DM}^{-1}$) predicted at zero time from the regression. Ruminal fluid space was calculated from the ratio of moisture to whole digesta sampled from the rumen at the fistula. Mean retention time was calculated from the concentration of marker (C_i) in ruminal digesta or feces, the median time of the sample (T_i), and the interval between subsequent samples (ΔT_i) as follows (Warner 1981):

$$\text{mean retention time (h)} = \frac{\sum(C_i T_i \Delta T_i)}{\sum C_i \Delta T_i}.$$

Digesta fill (kg DM) in the whole tract was calculated by the methods of Holleman and White (1989) from estimates of fecal output and mean retention time. Fecal output was calculated from measured food intake and digestibility of DM against lignin as follows:

$$\begin{aligned} \text{fecal output (g DM d}^{-1}\text{)} &= \text{DM intake (g d}^{-1}\text{)} \\ &\times (1 - \text{digestible fraction of DM}). \end{aligned}$$

Retention of DM in the whole tract was estimated as the sum of mean retention times for particles (Yb) in the rumen and the intestines. Indigestible fill was the product of fecal output and the average mean retention time for particles in each sea-

son. Total fill was based on indigestible fill and indigestibility against lignin with the assumption that assimilation of digesta DM declined exponentially with time in the digestive tract (Holleman and White 1989).

Ruminal degradation rates were calculated from the residual DM in the incubation bag. Residues of each dietary component such as N or cellulose were determined from the nutrient concentration in DM of pooled samples analyzed from each time point. Fractional rates of degradation ($-K; \text{d}^{-1}$) were calculated as the slope of the least squares linear regression of the natural logarithm of the residue against time in the rumen during spring (K_s), autumn (K_A), and winter (K_w).

We used programs in SYSTAT 11 (SYSTAT Software, Richmond, CA) to calculate and analyze statistics. Repeated measures of digesta DM or digestible fraction along the digestive tract were compared using ANOVA. Bacterial counts or pH were compared using ANOVA with season ($n = 3$) or time from feeding supplement and hay ($n = 2$) as a factor within animals. Seasonal measures of digesta passage were compared using ANOVA with season ($n = 3$) and dietary supplement ($n = 2$) nested within experimental trials ($n = 2$) as factors. Pairwise contrasts were performed with Bonferroni's adjustments for multiple comparisons. Data expressed as proportions or percentages (e.g., digestible fraction) were transformed to the arcsine of the square root to meet assumptions of normality for ANOVA (Zar 1999). Means are reported with 1 standard deviation ($\pm \text{SD}$). Statistical significance was determined as less than 5% probability of Type I error ($P < 0.05$).

Results

Diet

Hay was the principal source of ingested OM (92%), gross energy (92%), cellulose (97%), hemicellulose (100%), N (70%—

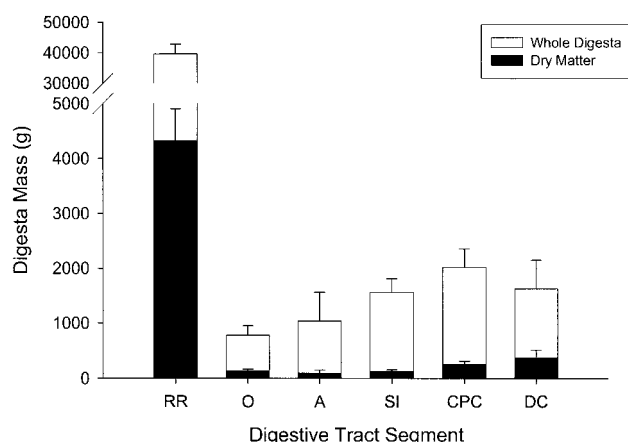


Figure 1. Distribution (mean $\pm$ SD) of whole (open bars) and dry (closed bars) mass of digesta along the digestive tract of castrated adult muskoxen ($n = 5$) in late winter (April): ruminoreticulum (RR), omasum (O), abomasum (A), small intestine (SI), cecum and proximal colon (CPC), distal colon (DC). Digesta fill in the ruminoreticulum was significantly greater ($P < 0.05$) than those of all subsequent segments. Segments after the ruminoreticulum did not differ in digesta fill ($P > 0.05$).

90%), and S (60%–84%) for muskoxen in spring and autumn (Peltier et al. 2003) and most likely accounted for the major fraction of the diet in winter for both experiments on segmental function and seasonal response. NDF consistently accounted for 61% of the DM in hay with more variation in the non-structural fractions of the diet, such as residual OM (20%–31%) and CP (5%–11%; Table 1). The low-fiber content of the supplements (24%–33%) provided muskoxen with an easily degraded source of CP (26%–13%) and nonstructural carbohydrate (34%–55% residual OM) that was also the principal source of Na during spring and autumn (Peltier et al. 2003). The diet of hay and supplement was adequate for maintenance of lean mass throughout the year even though digestible intakes of energy and N were lower in spring ($554 \pm 117 \text{ kJ kg}^{-0.75} \text{ d}^{-1}$ and $336 \pm 131 \text{ mg N kg}^{-0.75} \text{ d}^{-1}$) than in autumn (923 ± 242

$\text{kJ kg}^{-0.75} \text{ d}^{-1}$ and $768 \pm 252 \text{ mg N kg}^{-0.75} \text{ d}^{-1}$; Peltier et al. 2003).

Segmental Capacity and Function

Digestive tract contents were $16.0\% \pm 1.7\%$ of body mass ($292 \pm 14 \text{ kg}$) in late winter. Most of the digesta fill was associated with the foregut in the rumen and reticulum ($84.5\% \pm 2.0\%$) with only $3.3\% \pm 0.5\%$ in the small intestine, and $7.9\% \pm 2.0\%$ in the hindgut (Fig. 1).

Digestibility of hay over the whole digestive tract was estimated from rectal digesta at $0.54 \pm 0.03 \text{ g g}^{-1}$ of DM, $0.58 \pm 0.03 \text{ g g}^{-1}$ of NDF, $0.61 \pm 0.03 \text{ g g}^{-1}$ of hemicellulose, $0.62 \pm 0.03 \text{ g g}^{-1}$ of cellulose, $0.39 \pm 0.07 \text{ g g}^{-1}$ of CP, and $0.82 \pm 0.06 \text{ g g}^{-1}$ residual OM. Digestion in the reticulo-rumen accounted for 79% of DM ($0.43 \pm 0.05 \text{ g g}^{-1}$), 75% of NDF ($0.43 \pm 0.07 \text{ g g}^{-1}$), 69% of hemicellulose ($0.42 \pm 0.07 \text{ g g}^{-1}$), and 80% of cellulose ($0.50 \pm 0.09 \text{ g g}^{-1}$) degradation along the digestive tract. Ruminal digestion removed approximately 29% of apparently degradable CP ($0.12 \pm 0.18 \text{ g g}^{-1}$), whereas 97% of the degradable protein was removed along the small intestine ($0.38 \pm 0.13 \text{ g g}^{-1}$ at the proximal colon).

Seasonal Digesta Flow and Fill

Retention time was not affected by the type of supplemental ration fed to muskoxen in each season ($P > 0.05$). Fluid digesta flowed more quickly than particles through the rumen and the intestines (Table 2). The difference in mean retention between marked phases (2–3 h) in the rumen was similar across seasons, whereas differences in intestinal retention times were much more variable. Mean retention of particulate digesta did not vary with season in either the rumen or the intestines. Mean retention and turnover of fluids in the rumen varied by 10% between seasons (Fig. 2A).

Increases in food intake between spring and autumn were accompanied by gains in both the indigestible and digestible fractions of digesta in the tract (Table 3). Estimates of DM fill

Table 2: Mean retention times of fluid (Co-EDTA) and particulate (Yb-fiber) phases of marked digesta in the rumen and the intestines of castrated adult muskoxen

	Spring	Autumn	Winter	Statistic (P value)
Rumen (8 animals):				M (<.001)
Fluid	$9.6^{\text{AB}} \pm 1.41$ (16)	$8.7^{\text{A}} \pm 2.0$ (16)	$10.5^{\text{B}} \pm 1.4$ (16)	S (.016)
Particle	12.3 ± 3.5 (16)	12.3 ± 3.0 (16)	12.7 ± 2.9 (16)	S (.894)
Intestines, duodenum to rectum (4–6 animals):				M (.008)
Fluid	32.1 ± 20.3 (11)	28.1 ± 9.1 (9)	33.5 ± 13.9 (8)	S (.744)
Particle	29.4 ± 15.0 (9)	26.9 ± 10.3 (6)	36.8 ± 12.1 (8)	S (.255)

Note. Mean $\pm$ SD (sample size). Spring = May and June; autumn = August and September; winter = February and March. Statistical effects of marked phase (M) on each parameter and season (S) within a phase are listed with probability of the F statistic. Different superscripts indicate significant pairwise differences between seasons ($P < 0.05$).

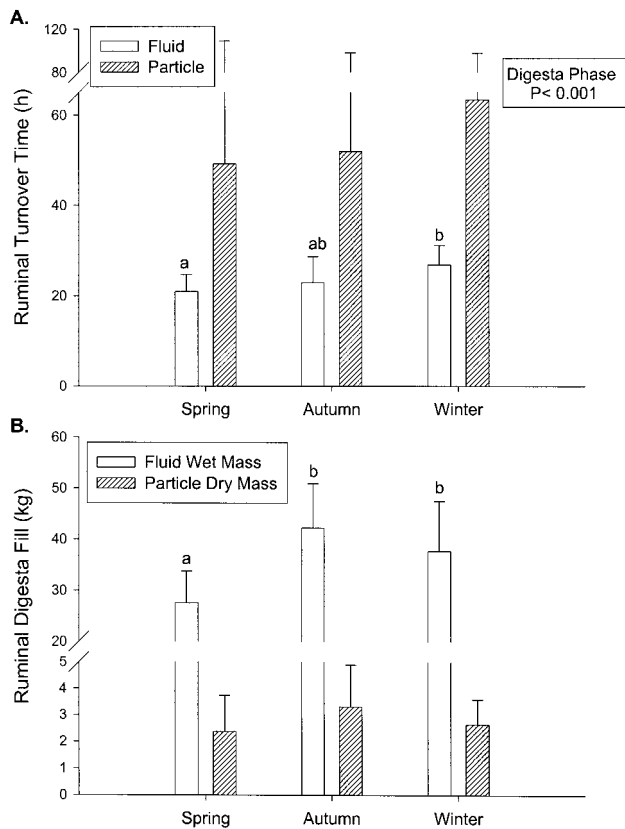


Figure 2. Ruminal turnover time (A) and fill (B) of fluid (Co-EDTA) and particulate (Yb-fiber) digesta in castrated adult muskoxen ($n = 8$) during spring (May and June), autumn (August and September), and winter (February and March; mean $\pm$ SD). Letters indicate significant differences ($P < 0.05$) between seasons within each phase of digesta.

were equivalent to 1.2% and 1.7% of body mass in spring and winter, respectively. Those estimates were similar to the fill of DM measured directly in muskoxen during late winter at $1.82\% \pm 0.2\%$ of body mass (Fig. 1). Gains in digesta fill during autumn hyperphagia were probably due to increases in the rumen because ruminal fluid space increased from 27.5 ± 6.3 kg in spring to 42.2 ± 8.7 kg in autumn (Fig. 2B). Ruminal fluid space was similar between autumn and winter (37.6 ± 9.9 kg) and significantly greater than the fluid space in spring (Fig. 2B). Estimated digesta fill from markers was $15.8\% \pm 4.3\%$ and $1.08\% \pm 0.5\%$ of body mass for fluid wet mass and particle dry mass, respectively (Fig. 3B). Marker-based estimates were similar to direct measures of ruminal digesta fill in muskoxen killed in late winter for studies of segmental function (see "Segmental Capacity and Function"): 35.2 ± 3.0 kg water or $12.1\% \pm 1.5\%$ body mass; 4.3 ± 0.5 kg DM or $1.5\% \pm 0.2\%$ body mass.

Seasonal Rumen Function

Ruminal degradation was measured for hay, the principal component of the diet in each period, and for two supplements. Fractional disappearance of DM from the incubated bags fitted a log-linear relationship with time for both hay ($R^2 = 94.0\% \pm 8.3\%$ for 72 regressions) and supplements ($R^2 = 93.1\% \pm 7.3\%$ for 58 regressions).

Degradation of hay was not affected by the supplements consumed in each trial ($P > 0.05$). Hay DM disappeared from the incubation bags most rapidly during hyperphagia in autumn than during spring and winter leaving much smaller residues after both 24 and 48 h in the rumen (Fig. 3A). Supplements were more rapidly degraded than the hay in all seasons, but fractional rates of DM degradation were greater for the basic supplement than for the protein supplement (Fig. 3B, 3C). The disappearance of supplement DM also varied with season and left the smallest residues of DM in autumn after 48 h of incubation (Fig. 3B, 3C).

Ruminal disappearance of C, cellulose, and hemicellulose from incubated hay followed the same seasonal pattern as DM, with the fastest rates of loss in autumn (Fig. 4). Appearance of N on ruminal residues of hay in spring and winter reflected the relatively low N content of the forage (Table 1) as well as the larger residue of DM at the end of the incubation. Greater degradability and higher concentrations of N in supplements resulted in net release of N to the rumen in all seasons (Fig. 5). Microbial actions on supplements followed the seasonal pattern for hay; that is, release of C, N, S, and ash from supplements was generally greater in autumn than spring (Fig. 5).

Bacterial numbers in rumen fluid reflected the seasonal rates of ruminal degradation of both hay and supplements that was highest in autumn during hyperphagia. Bacterial numbers increased by 30% between spring and autumn but declined by 49% to winter (Table 4). Consumption of the daily ration of fermentable supplement with any fresh hay did not alter bacterial densities within 6 h in any season. Ruminal acidity was greatest in autumn hyperphagia and further enhanced by 0.2 ± 0.2 units to pH 6.6 ± 0.1 within 6 h of feeding in each season (Table 4).

The total concentration of fermentation acids reflected changes in ruminal pH between seasons. Ruminal acid concentrations were not affected ($P > 0.05$) by the type of supplement either before or after feeding. Prefeeding concentration of total SCFA increased in autumn (128 ± 12 mmol L^{-1}) by 16% above the concentrations in spring and winter (109 ± 15 mmol L^{-1} ; Fig. 6A). Acetate was the principal acid in each season at 73%–78% of the total molar proportion. Concentrations of propionate and butyrate also increased in autumn by 25% and 46%, respectively, in comparison with spring and winter. Consequently, molar proportions were greater in autumn than in either winter or spring for both propionate ($15.7\% \pm 1.1\%$ vs. $14.5\% \pm 0.9\%$) and butyrate ($8.4\% \pm 1.6\%$

Table 3: Intake, digestibility, fecal output, and fill of dry matter (DM) in castrated adult muskoxen ($n = 8$)

	Spring	Autumn	Winter
Body mass (kg)	239 ^A $\pm$ 39	260 ^B $\pm$ 37	269 ^B $\pm$ 30
Measured intake (kg d ⁻¹)	2.75 ^A $\pm$.54	4.75 ^B $\pm$.74	ND
Digestibility (g g ⁻¹)	.691 ^A $\pm$.086	.711 ^A $\pm$.084	.501 ^B $\pm$.067
Fecal output (kg d ⁻¹)	.86 ^A $\pm$.27	1.38 ^B $\pm$.51	ND
Total mean retention time of particles (h) ^a	41.70	39.20	49.50
Indigestible fill (kg)	1.49 ^A $\pm$.47	2.26 ^B $\pm$.84	ND
Total fill (kg)	2.80 ^A $\pm$.63	4.42 ^B $\pm$.94	ND

Note. Mean $\pm$ SD. Spring = May and June; autumn = August and September; winter = February and March. Different superscripts indicate significant differences between seasons ($P < 0.05$). ND = not determined.

^a Average retention of DM in the whole tract calculated as the sum of mean retention times for particles in the rumen and the intestines (see Table 2).

vs. $6.6\% \pm 0.9\%$). Minor acids also increased in concentration (Fig. 6B) and molar proportion between spring and autumn (1.70 ± 0.24 vs. 2.94 ± 0.64 mmol L⁻¹; 1.63 ± 0.27 vs. $2.30\% \pm 0.44\%$) but did not change from autumn to winter (2.73 ± 0.50 mmol L⁻¹; $2.43\% \pm 0.45\%$).

Increases in total SCFA concentration after feeding changed with season; postprandial increment of total SCFA was only 2% in autumn, whereas postprandial gains in spring and winter were much greater at 13%. Seasonal differences in the postprandial concentrations of acetate, propionate, and butyrate were therefore less pronounced than those before feeding (Fig. 6A, 6C). The seasonal pattern of concentrations among the minor SCFA was conserved after feeding supplement and hay. Feeding depressed concentrations of isobutyrate and isovalerate but enhanced those of valerate and caproate, especially during winter and spring.

Discussion

Hyperphagia and hypophagia can be defined as the relationship between food intake and the requirements of nutrients and energy to maintain the body (Barboza and Hume 2006). Autumn is a period of hyperphagia in muskoxen because high food intakes accompany deposition of excess nutrients and energy in body mass. Conversely, low food intakes with slow loss of body mass in muskoxen indicate deficits of energy and nutrients or hypophagia from late winter through spring (Peltier et al. 2003). Food intakes of castrates in midwinter were probably as low as those in spring (Table 3). Muskoxen may, however, maintain body mass in midwinter because basal energy requirements are lower in winter than summer (Lawler and White 1997). The relationship between ad lib. food intake and requirements in winter may therefore depend on the size of body stores and additional energy demands for locomotion and thermoregulation in nonreproductive animals (Peltier et al. 2003; Barboza and Hume 2006). Cycles of hyperphagia and

hypophagia in muskoxen probably follow a common seasonal cue from photoperiod because the patterns for castrated males are similar to those of immature animals (Peltier and Barboza 2003) and adult females (Adamczewski et al. 1994a; Rombach et al. 2002b).

The ability to respond to increasing food intake requires flexibility in digestive function and capacity. The pattern of digestion in captive muskoxen was similar to wild muskoxen and to domestic species of grazing ruminants such as cattle and sheep (Staal and Thing 1991). Ruminal fermentation accounted for most of the digestive capacity (Fig. 1) and DM digestion in muskoxen. Most dietary DM was associated with structural carbohydrates such as hemicellulose and cellulose that were digested in the rumen. Conversely, postruminal segments of the digestive tract were more important than the rumen (Fig. 4) for net digestion of N in CP.

Sustaining Hyperphagia

Muskoxen sustained hyperphagia by conserving the time available for digestion and absorption. Ruminal retention of digesta was maintained within 10% (Table 2) even though food intake increased by 74% between spring and autumn. Intestinal retention was likewise conserved between seasons (Table 2), which provided a similar amount of time for digestion and absorption in the small intestine and hindgut. Consistent differences in retention of fluid and particulate digesta in the rumen indicated that patterns of ruminal mixing did not change with DM intake between seasons. Hyperphagia probably did not alter the proportion of postruminal digestion in muskoxen because intestinal passage also did not change between seasons. Similar changes in intake of fibrous diets may not alter the site of DM digestion in domestic grazers such as cattle and sheep (Galyean and Owens 1991).

The ability to conserve time for digestive processes in mus-

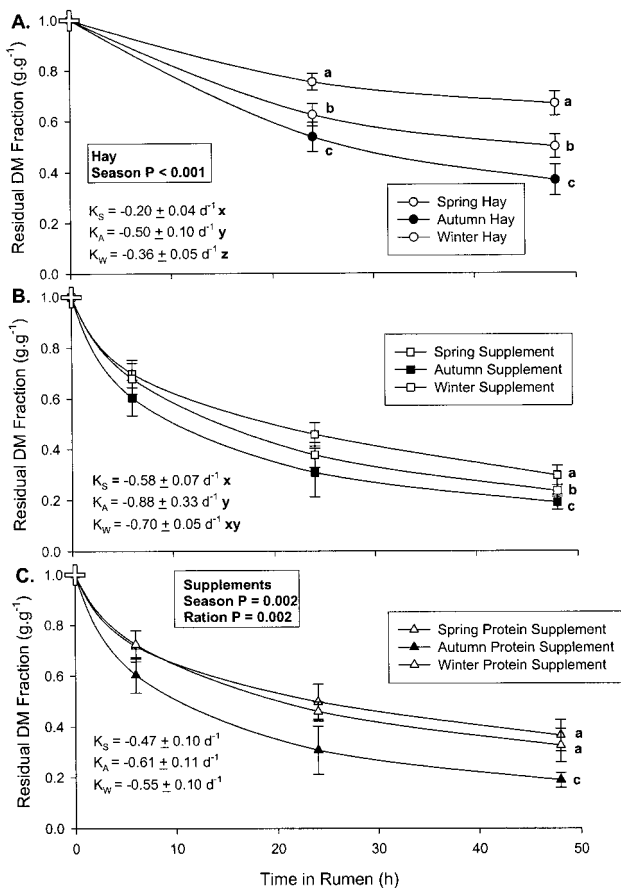


Figure 3. Residual fraction of dry matter (DM) from hay (A), supplement (B), and protein supplement (C) incubated in the rumen of castrated adult muskoxen ($n = 8$) during spring (May and June), autumn (August and September), and winter (February and March; mean $\pm$ SD). Fractional degradation rates (K) are the slopes from the regression of $\ln$ (residual DM fraction) against time in the rumen during spring (K_S), autumn (K_A), and winter (K_W). Different letters indicate significant differences between seasons ($P < 0.05$).

koxen was probably due to a gain in digesta fill of 58% as food intake increased between spring and autumn (Table 3). The range of digesta fill in muskoxen may be greater than other herbivores because total digesta fill of castrates in late winter was $24\% \pm 13\%$ greater than the alimentary fill predicted from body mass (Demment and Van Soest 1985; White et al. 1987). This suggestion is supported by measures of digesta fill in wild muskoxen that are 20%–28% of body mass (Staal and Thing 1991; Adamczewski et al. 1995) and greater than those of captive animals ($16.0\% \pm 1.7\%$). The large digestive capacity of muskoxen is consistent with their grazing habits because ruminal digesta fill of captive muskoxen in late winter ($13.61\% \pm 1.65\%$ body mass) was similar to that predicted by an allometric relationship for grazing ruminants (13.7% body mass; Clauss et al. 2003a). The ability to mix the large volume

of fibrous digesta is reflected in thick ruminal muscles and mucosal cornification in muskoxen and other grazers (Clauss et al. 2003b; Knott et al. 2004).

Large digestive capacity in muskoxen facilitates the accommodation of less digestible forage. Consistent ruminal fill of both particulate and fluid digesta between autumn and winter (Fig. 2B) reflects the low digestibility of DM (Table 3) and the slow degradation of foods in the rumen (Fig. 3) during winter. Consequently, ruminal digestibility could limit food intakes of muskoxen in autumn when ruminal fill is at a maximum. Ruminal rates of degradation decline with concentrations of structural carbohydrates and lignin because hay was more slowly degraded than supplement (Table 1; Fig. 3). Senescent sedge and moss may therefore occupy more ruminal space and constrain food intake more than less fibrous forbs and browse. Gains of body mass in muskoxen may be limited by both the abundance and quality of forages in late autumn. This suggestion requires confirmation from studies of fluid and particle kinetics in muskoxen fed different foods during hyperphagia.

Differential passage of fluids and particles in the rumen was not as pronounced in muskoxen (particle/fluid mean retention = 1.31 ± 0.35 ; Table 2) as in other grazers (1.56 to 3.80 ; Clauss and Lechner-Doll 2001). A relatively rapid flow of particles could indicate that ruminal outflow constrains food intake in muskoxen less than other ruminant grazers when animals attain maximum ruminal fill during hyperphagia. Ruminal motility may also provide more consistent mixing and retention than the lower tract in muskoxen throughout the year. Consequently, retention times of fluids and particles were shorter and less variable in the rumen than in the intestines in muskoxen (Table 2). This pattern of motility may be similar among grazers at high latitudes because postruminal retention of Cr-mordanted fiber changes more than ruminal retention as food intake changes with season in bison (*Bison bison*; Rutley and Hudson 2000). Estimates of retention also reflect the physical characteristics of the labeled particles: total mean retention times of Yb-fiber (75 – $500 \mu\text{m}$) in this study (39 – 50 h; Table 3) are shorter than those reported for long hay mordanted with Cr in muskoxen (95 – 115 h; Adamczewski et al. 1994a). The effect of seasonal changes in food intake on particle kinetics requires further study with multiple markers to relate particle size and specific gravity to rumination and passage rate (Faichney 1986; Ulyatt et al. 1986; Gross et al. 1995; Ellis et al. 1999) in muskoxen and other northern grazers.

Concomitant increases in microbial degradation and food intake in muskoxen (Figs. 3–5) enhance digestive capacity by increasing the digestible component of digesta fill (Table 3). Faster forage degradation between spring and autumn was associated with a gain in the total number of microbes in the rumen. Bacteria in the fluid phase of the rumen probably increased by 89% from spring to autumn because of the combined effects of a 45% gain in fluid volume (Fig. 2B) and a 30% gain in the number of bacteria (Table 4). The largest

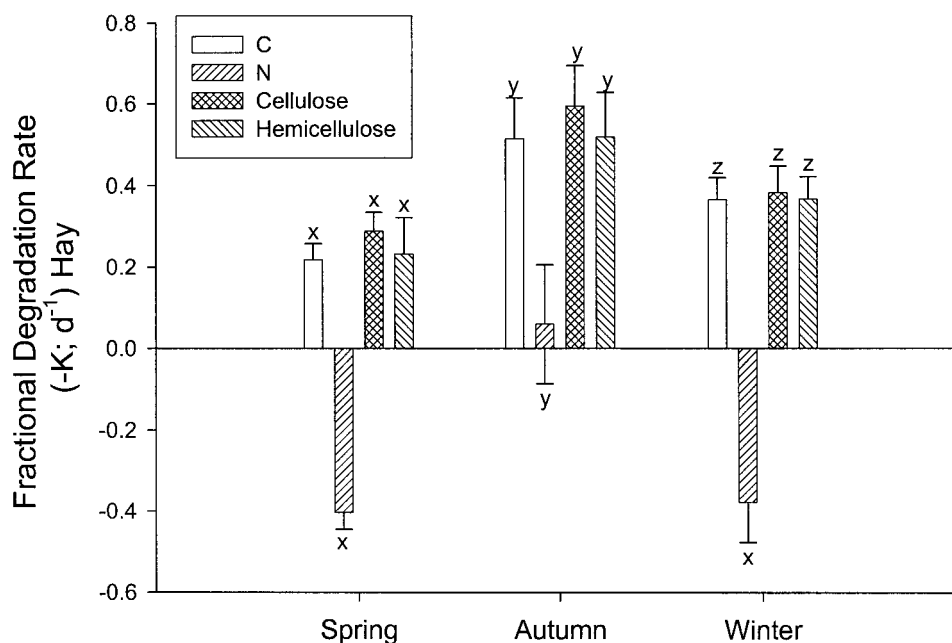


Figure 4. Fractional degradation rates of C, N, and structural carbohydrate in hay incubated in the rumen of castrated adult muskoxen ($n = 8$) during spring (May and June), autumn (August and September), and winter (February and March; mean $\pm$ SD). Different letters indicate significant differences between seasons within each component ($P < 0.05$).

fraction of the bacterial population are those organisms associated with the particulate phase of digesta that are directly involved with fiber degradation (Olubobokun et al. 1988; Olubobokun and Craig 1990; McAllister et al. 1994). Microbial attachment to forages in muskoxen was reflected in the apparent gain of N in incubated hay (Fig. 4) during winter and spring when residual DM was greater than in autumn. Bacterial populations associated with particles probably increased in a similar fashion to the fluid population as total digesta fill increased between spring and autumn. Distribution of bacteria between fluid and particulate phases do not change with rumen fill in sheep if the physical characteristics of the digesta are unchanged (Rodríguez et al. 2003). This suggestion awaits further measures of bacterial production in muskoxen that include both fluid and particle associated fractions in each season.

Fiber digestion in the rumen is the result of a community rather than one superorganism or rate-limiting enzyme (McAllister et al. 1994) because the biochemical activities of ruminal microbes are interconnected (Theodorou and France 1993). Colonization of forage particles is initiated by fibrolytic bacteria such as *Ruminococcus albus*, *Ruminococcus flavefaciens*, and *Fibrobacter succinogenes*, which complement fungal invasions of the cell wall matrix (Gordon and Phillips 1998). Subsequent attachment of bacteria with complementary enzymes establishes a biofilm within which substrates are exchanged to produce ultimately the SCFA absorbed by the host. High rates of in situ degradation during hyperphagia are probably not a sim-

ple outcome of increased microbial counts in muskoxen even though more bacteria could increase rates of colonization on food residues and shorten the lag time in beginning digestion of substrates (Spalinger et al. 1986; Ellis et al. 1999). Induction of attachment systems may be crucial in switching to unusual forages such as lichens (Olsen and Mathiesen 1998), but seasonal differences in those delays are probably small when diet changes are small as it was for our studies of captive muskoxen.

Ruminal rates of degradation (K) doubled between spring and autumn for N in supplements (Fig. 5) and for cellulose and hemicellulose in hay (Fig. 4). Changes in degradation kinetics suggest seasonal differences in the specificity of attachment and in the maximum velocity of enzymes for degradation of structural carbohydrates and protein. Those kinetic changes are probably due to expression of different enzymes either by the same community of bacteria-exchanging genes (Woese 2002) or by a shift in the composition of the community that resulted in the expression of new genes (Dale and Park 2004). Changes in the diversity of the microbial community can accompany dietary changes in domestic ruminants (Kocherginskaya et al. 2001). It is likely that the gradual seasonal transition between intakes of food in muskoxen favors a shift in community structure by altering the numbers of each phylotype in the population (Stahl et al. 1988; Tajima et al. 2001). This hypothesis requires testing by seasonal measures of bacterial diversity in both liquid and particulate fractions of digesta and

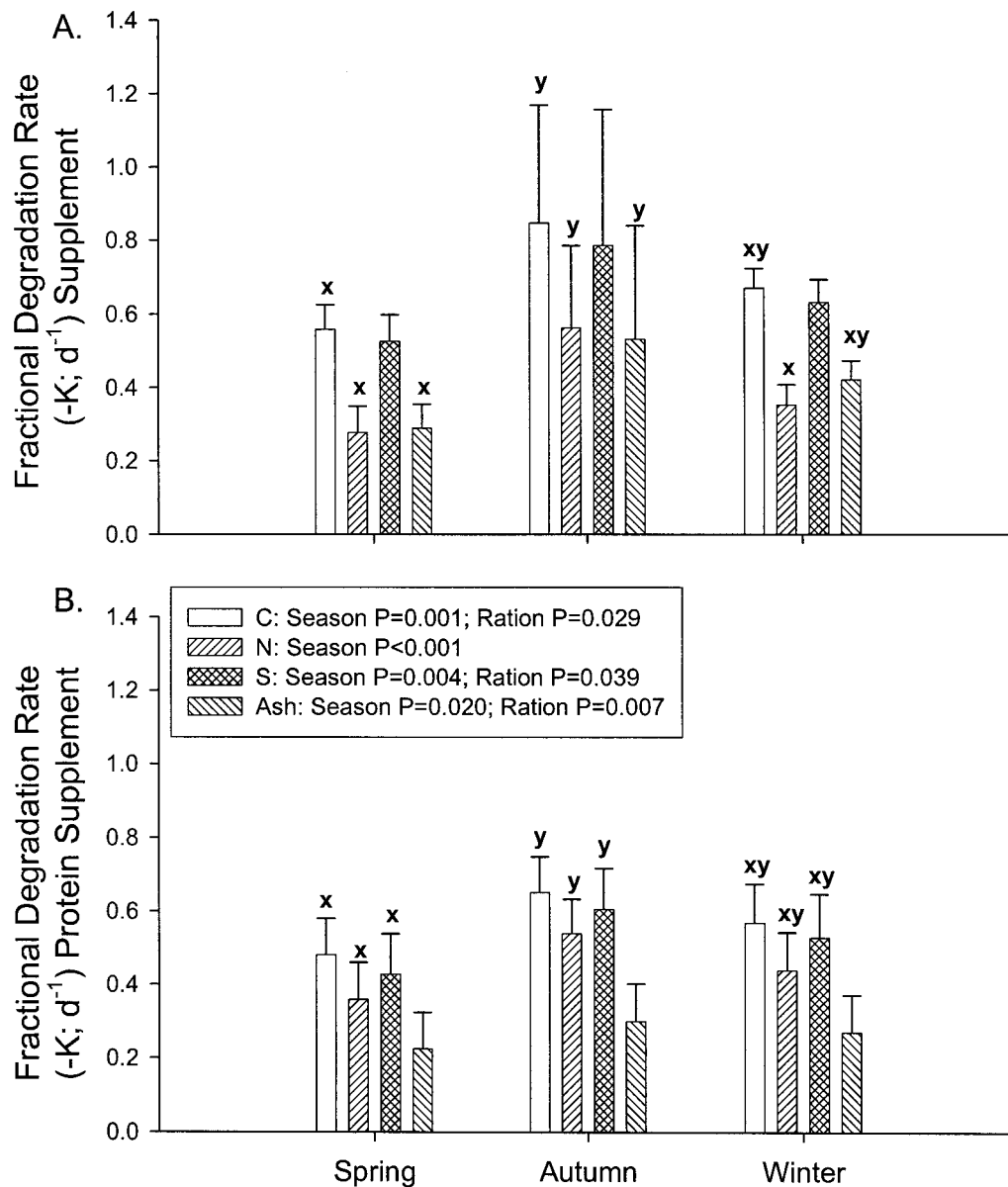


Figure 5. Fractional degradation rates of C, N, S, and ash from two supplements incubated in the rumen of castrated adult muskoxen ($n = 8$) during spring (May and June), autumn (August and September), and winter (February and March; mean $\pm$ SD). Different letters indicate significant differences between seasons within each component ($P < 0.05$).

by counts of phylotypes with characteristic functions such as fiber degradation.

Seasonal hyperphagia increased total SCFA concentration by only 16% even though SCFA production probably increased in concert with the doubling of degradation rates for fiber (Fig. 4). This disparity between increments in rates of degradation and SCFA concentration probably reflects gains in absorption rates of SCFA that would minimize accumulation of acids in the rumen (Rechkemmer et al. 1995). This suggestion is supported by changes in butyrate uptake by Na^+/H^+ exchange that

are greater during summer hyperphagia than winter in the rumen of wild and captive reindeer (Storeheier et al. 2003). The narrow range of ruminal pH in muskoxen (Table 4) also indicates that secretion of buffers and absorption of acid match degradation rates. Hyperphagia in muskoxen is therefore not accompanied by acidotic conditions that can suppress cellulolytic bacteria in domestic ruminants (Russell and Wilson 1996). Small decreases in ruminal pH after feeding are consistent with the corresponding gains in SCFA in muskoxen and are similar to the responses of domestic ruminants to diurnal

Table 4: Concentrations of bacteria and pH in ruminal fluid before feeding supplement and 6 h after feeding supplement and fresh grass hay to muskoxen ($n = 8$)

	Spring	Autumn	Winter	Statistic (P value)
Bacterial number ($\times 10^9 \text{ mL}^{-1}$):				F (.698)
Before feeding	3.51 ^A $\pm$.45	4.55 ^B $\pm$.52	2.34 ^C $\pm$.67	S (<.001)
After feeding	3.47 ^A $\pm$.52	4.51 ^B $\pm$.37	2.51 ^C $\pm$.24	S (<.001)
Digesta pH:				F (.001)
Before feeding	6.8 ^{AB} $\pm$.2	6.7 ^B $\pm$.1	6.9 ^A $\pm$.2	S (.029)
After feeding	6.5 $\pm$.2	6.6 $\pm$.1	6.7 $\pm$.1	S (.052)

Note. Mean $\pm$ SD. Spring = May and June; autumn = August and September; winter = February and March. Different superscripts indicate significant pairwise differences between seasons ($P < 0.05$). Statistical effects of feeding (F) on each parameter and season (S) are listed with probability of the F statistic.

patterns of forage consumption and to small doses of fermentable supplements (Dehority 2003).

Seasonal changes in the concentration of the major SCFA in muskoxen suggest changes in the microbial pathways of energy flow during hyperphagia that may alter the efficiency of microbial growth as well as the return of energy to the host (Russell and Strobel 1993). SCFA production is mainly due to carbohydrate fermentation in the rumen (Fig. 2), which provides $90\% \pm 3\%$ of the digestible energy of muskoxen in spring and autumn if energy intakes (Peltier et al. 2003) are divided between CP (gelatins 22.05 kJ g^{-1}), hemicellulose (17.82 kJ g^{-1}), cellulose (17.49 kJ g^{-1}), and residual OM (starch 17.48 kJ g^{-1} ; Blaxter 1989). Most of the energy is probably absorbed as acetate, the largest molar proportion of the SCFA, which would have been easily deposited as body lipid that increased by 23% in castrated muskoxen during autumn (Peltier et al. 2003). Increases in the molar proportion of propionate during autumn could also provide glucogenic substrates that would support maintenance of lean mass in castrates. Similar gains in propionate and butyrate supplies for females would also support milk production and minimize their loss of lean and fat tissue during the brief summer (Parker et al. 1990; White et al. 1997). Direct measures of SCFA production and absorption are required to test these hypotheses of seasonal changes in fermentation yield to muskoxen.

The presence of minor SCFA such as isobutyrate, isovalerate, valerate, and caproate in rumen fluid suggests that small amounts of protein are fermented in the rumen of muskoxen (Bohnert et al. 2002b). Protein in supplements were degraded at similar rates (Fig. 5) even though their concentrations of N differed by a factor of 2 (Table 1). The composition of food proteins probably affected microbial degradation because S from fishmeal in the protein supplement was removed more slowly than from the basic supplement (Fig. 5). The lack of any difference in minor SCFA concentration between consumption of the basic and high-protein supplements is consistent with the greater importance of hay that supplied 83%–90% of dietary N to muskoxen (Peltier et al. 2003). The flow of dietary proteins to the small intestine depends on their sol-

ubility and retention in the rumen as well as their fermentability in each season (Fig. 5; Sauvant and van Milgen 1995). It is likely that most organic N is absorbed as microbial amino acids from the small intestine of muskoxen because seasonal changes in ruminal fill and degradation (Figs. 2, 3) would increase the ruminal efflux of microbes in a pattern similar to domestic ruminants (Poncet et al. 1995; Bohnert et al. 2002a). Changes in the corresponding postruminal load of amino N between spring and autumn would increase the demands for proteolysis and absorption by more than 135% (Peltier et al. 2003) in muskoxen because of gains in the flow of dietary, microbial, and endogenous protein. The abomasum and duodenum of muskoxen are well developed from an early age (Knott et al. 2004, 2005), which suggests that muskoxen can probably process these large nutrient loads as well as many domestic ruminants selected for high food intakes and mass gain (Robinson et al. 1999; Satter et al. 1999).

Consequences of Hypophagia

The maintenance of ruminal fill in muskoxen from autumn to winter (Fig. 2B) indicates a large pool of substrates were apparently available when bacteria declined to the lowest number in rumen fluid (Table 4). Winter declines in ruminal bacteria may be due to limiting nutrient supply or poor conditions for microbial growth. It is unlikely that N supply limits microbial growth in winter because N intakes were adequate to maintain lean mass of the host and because the ratio of digestible N to energy in spring exceeded the maximum theoretical yield for microbial growth in domestic ruminants (1.64 vs. $1.33 \text{ mg N kJ}^{-1}$; Nolan 1993). Low ruminal substrate loads in spring and winter may be inadequate for the most active phylotypes in the microbial community that are most abundant in autumn. The decline in bacterial numbers could therefore reflect the diminution of those active phylotypes. Large postprandial gains in concentration of the major SCFA during winter and spring in muskoxen (Fig. 6) may suggest that bacterial populations are energy limited compared with autumn. This suggestion awaits measures of SCFA absorption because postprandial accumu-

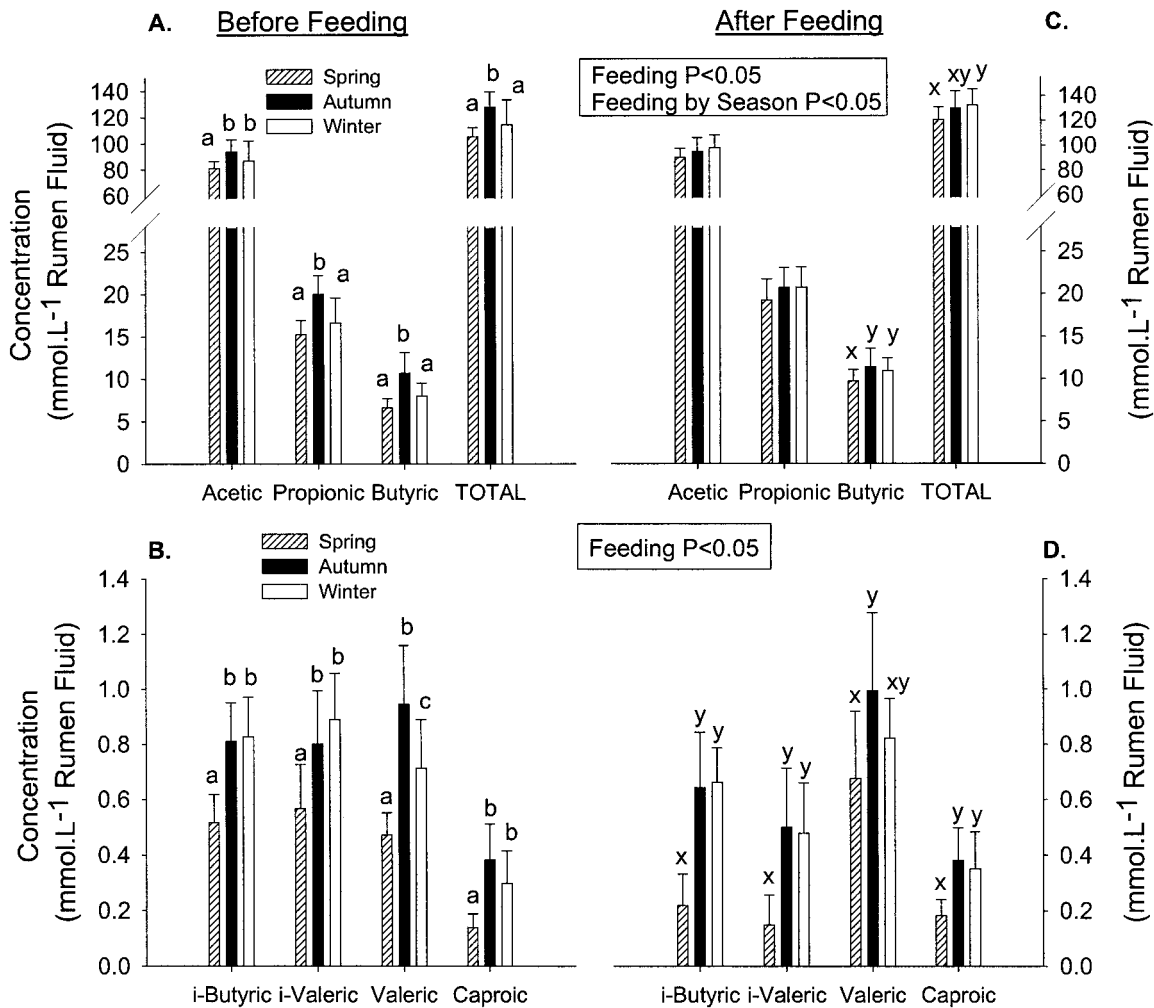


Figure 6. Concentrations of major (A, C) and minor (B, D) fermentation acids in the rumen of castrated adult muskoxen ($n = 8$) during spring (May and June), autumn (August and September), and winter (February and March; mean $\pm$ SD). Measures were taken before (A, B) and 6 h after (C, D) feeding supplement and fresh hay in each season. Different letters indicate significant differences between seasons within each acid ($P < 0.05$).

lation of SCFA may be also an outcome of seasonal reductions in absorption of SCFA (Storeheier et al. 2003). The most active microbes may be suppressed in muskoxen during winter if ruminal conditions of pH, SCFA, and osmolality become more variable because of changes in feeding frequency, absorption, and thermoregulation. Less frequent feeding in muskoxen during winter (Forchhammer 1995) may increase variation in ruminal pH and SCFA, as reported for domestic ruminants (Kaufmann et al. 1980; Soto-Navarro et al. 2000). Muskoxen may also reduce the effort of regulating ruminal conditions during winter because the endogenous heat increment after consumption of a standard meal declines between summer and winter; that is, muskoxen spend less energy on processing a meal in winter than in summer (Lawler and White 2003). Furthermore, ingestion of cold fluids and reductions of ruminal temperatures

(Cunningham et al. 1964) may also reduce microbial populations and activities in muskoxen during winter. Fermentative processes such as ureolysis and fiber adhesion are impaired at low temperature (Olubobokun et al. 1988; Roger et al. 1990; Barboza et al. 1997). Our ability to resolve these hypotheses requires further information on the thermal tolerance and substrate demands of the microbial community in each season.

Ecological Consequences

Declines in microbial activity during winter may be advantageous when food abundance and quality are low. Muskoxen still retain a large digestive capacity in winter that can accommodate senescent forage if sufficient food is available to support any short-term increases in energy demands. The energetic

costs of sustaining the most active fermentation would be best supported when food is predictably abundant and when the net energy can be deposited in body mass during autumn.

Consistent patterns of seasonal degradation among hay and supplements indicate that foods such as woody browse, which contain more nonstructural carbohydrate and protein than graminoids, are also fermented more rapidly during seasonal hyperphagia in muskoxen. The ability to process all forages more rapidly and completely during hyperphagia is consistent with rapid mass gains of muskoxen during summer and with rapid growth of muskox populations introduced to new ranges (Le Henaff and Crete 1989; Reynolds 1998). The ability to extract energy from low-quality forages may also minimize the area required for foraging and is consistent with the small home range of muskox herds (Reynolds et al. 2002; Barboza and Reynolds 2004).

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